

Matching Starbursts



The game is simple. Reach your hand into the bag and choose a Starburst. Replace the Starburst into the bag, mix up the bag, then reach in for another. If the color matches the first one, you win! What is the probability of winning this game?

1. Your group should play the game 10 times total.
   a. Number of wins for the group =
   b. Based on the results from your group, what is the probability of winning this game?
2. Let's see what happened in the other groups. Record your number of wins at the front of the room.
   a. Number of wins for the class =
   b. Based on the results from your whole class, what is the probability of winning this game?
3. How could we improve this estimate of the probability of winning?



4. To determine the true probability of winning, we can list out all possible pairs of choices that we could get. Complete the table.

1st candy \ 2nd candy						

5. Complete the following:
   a. Circle the outcomes with both red. What is the probability of winning with two red?
   b. Circle the outcomes with both yellow. What is the probability of winning with two yellow?
   c. Overall, what is the probability of winning?

Lesson 3.3 – Rules for Probability

QuickNotes

Check Your Understanding

Pew Research Center reported that family size is shrinking. Suppose we are to randomly select one mother and record the number of children. Here is a probability model.

Number of Children	1	2	3	4+
Probability	0.22	?	0.24	0.13

(a) Steven argues that the probability of a mother having exactly 2 children is ¼ because there are four categories (1,2,3,4+). Explain why this is incorrect.

(b) Find the probability of a mother having exactly 2 children. Show your work.

For each of the following write the event using proper notation and find the probability:

(c) Find the probability that a randomly selected mother has less than 2 children.

(d) Find the probability that a randomly selected mother has 1 or 2 children.

(e) Find the probability that a randomly selected mother does not have 4 or more children.